

Auteur: Noran Ben Said  
 Studierichting: Informatica  
 Oefeningenreeks 6B nr. 17.7

$$1^3 + 3^3 + 5^3 + \dots + (2n-1)^3$$

$$s_n = \sum_{k=1}^n (2k-1)^3$$

$$(2k-1)^3 = (2k)^3 - 3(2k)^2(1) + 3(2k)(1)^2 - 1^3$$

$$= 8k^3 - 12k^2 + 6k - 1$$

$$s_n = \sum_{k=1}^n (8k^3 - 12k^2 + 6k - 1)$$

$$= 8 \sum_{k=1}^n k^3 - 12 \sum_{k=1}^n k^2 + 6 \sum_{k=1}^n k - \sum_{k=1}^n 1$$

$$\sum_{k=1}^n k^3 = \left( \frac{n(n+1)}{2} \right)^2 = \frac{n^2(n+1)^2}{4}$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n 1 = n$$

$$s_n = 8 \cdot \frac{n^2(n+1)^2}{4} - 12 \cdot \frac{n(n+1)(2n+1)}{6} + 6 \cdot \frac{n(n+1)}{2} - n$$

$$= 2n^2(n+1)^2 - 2n(n+1)(2n+1) + 3n(n+1) - n$$

$$= 2n^2(n^2 + 2n + 1) - 2n(n+1)(2n+1) + 3n(n+1) - n$$

$$= 2n^4 + 4n^3 + 2n^2 - 2n(2n^2 + 3n + 1) + 3n^2 + 3n - n$$

$$= 2n^4 + 4n^3 + 2n^2 - 4n^3 - 6n^2 - 2n + 3n^2 + 2n$$

$$= 2n^4 - n^2$$

$$= n^2(2n^2 - 1)$$

$$s_n = n^2(2n^2 - 1) \text{ voor alle positieve gehele getallen } n.$$

## References